

Understanding ISO 11171: The NIST Traceable Particle Counter Calibration Standard

Automatic liquid particle counters (LPC) are used to monitor contamination levels in hydraulic oil to determine component and assembly cleanliness levels, filter efficiencies and size ratings. Prior to 1999, liquid particle counter calibration for hydraulic applications was done using ISO 4402 and AC Fine Test Dust (ACFTD) with recognized shortcomings.

With the passage of ISO 11171:1999, a National Institute of Standards and Technology ([NIST](#)) traceable method of calibration including a statistical method for the procedure, sample prep and traceability in contamination control was achieved, saving valuable operator time while generating more accurate and repeatable performance data. Liquid particle counter accuracy and reliability has also improved making them very useful in a variety of industries including aerospace, hydraulics, power generation and the military.

Calibration Changes, Impacts and Challenges

ISO 11171:1999 established baseline liquid particle counter performance specifications for volume measurement reproducibility, resolution and counting accuracy including:

- NIST-certified calibration suspensions (SRM 2806)
- Statistical methods for validating analytical procedures
- Minimal instrument performance requirements & operating limits
- Critical samples analysis standard (in triplicate with multiple counts for each sample to uncover systemic errors in trend data)
- [Threshold noise level](#), coincidence error & [flow rate](#) limits of the LPC must be determined

ISO MTD (Medium Test Dust) replaced ACFTD (Air Cleaner Fine Test Dust) in fluid suspensions used to calibrate optical particle counters. Although the new primary calibration fluid (SRM 2806) was more suitably certified for particle counter calibration and NIST traceable, calibration standards using [SRM 2806](#) generated different calibration data than ACFTD-based fluids, resulting in higher contamination level results for samples measured by instruments calibrated with the MTD-based fluid.

As a result, the ISO 4406 reporting standard was revised to include a three-tier cleanliness code outlining counts for particles > 4µm(c), 6µm(c) and 14µm(c) to generate the code while leaving the actual code table unchanged. Such measurements can be performed with ISO 11171-calibrated optical particle counters such as the [HIAC 8011+](#) lab particle counter and [PODS+](#) portable particle counter. These modifications minimized changes in count data and cleanliness codes on oil analysis reports but required the previously reported sizes (2 µm, 5 µm and 15µm) be replaced with 4 µm(c), 6 µm(c) and 14 µm(c).

In mid-2014, NIST released a third batch of certified primary calibration fluid—SRM 2806b. The counts for this new standard were notably higher than prior batches due to the higher concentration of test dust used and a more accurate certification process. As the new MTD content was not published, the missing information made it difficult to quantify the impact of the higher concentration and more accurate certification

Real-world sampling shows a considerable increase in particle counts when samples are tested on particle counters with SRM 2806b traceable calibrations. As a result, fluid samples ‘appear’ dirtier and filters ‘less efficient’. The results are different because they are based on two different ISO particle count calibrations.

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[USP<643>](#)

➤ [21 CFR Part 11: Data Integrity and Compliance](#)

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▼ ISO 11171 Standard

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